## Flight Compartment Windshields – General Data

1. General

Basic pilot's and co-pilot's windshields Mod 6/2032 embodies 3/8 inch thick acrylic plastic, bird impact resistant windshields.

Mod S.O.O. 6187 Mod 6/2045 embodies electrically heated, bird impact resistant windshields manufactured from laminates of glass and vinyl. Sensing and heating elements are incorporated within the laminations during manufacture.

Windshield heating control is achieved through a windshield heat control switch, circuit breakers, two temperature controllers, a heat control relay and associated wiring.

Both windshields are secured in the flight compartment windshield frames, against retaining angles, by retaining plates attached by screws. Sealing strips are located each side of the windshield between retaining angles and plates. During installation, the space between the edge of the windshield and the frame structure is filled with sealant (refer to 20-50-01, Weather Sealing – General Data) to prevent entry of moisture.

2. Removal/Installation

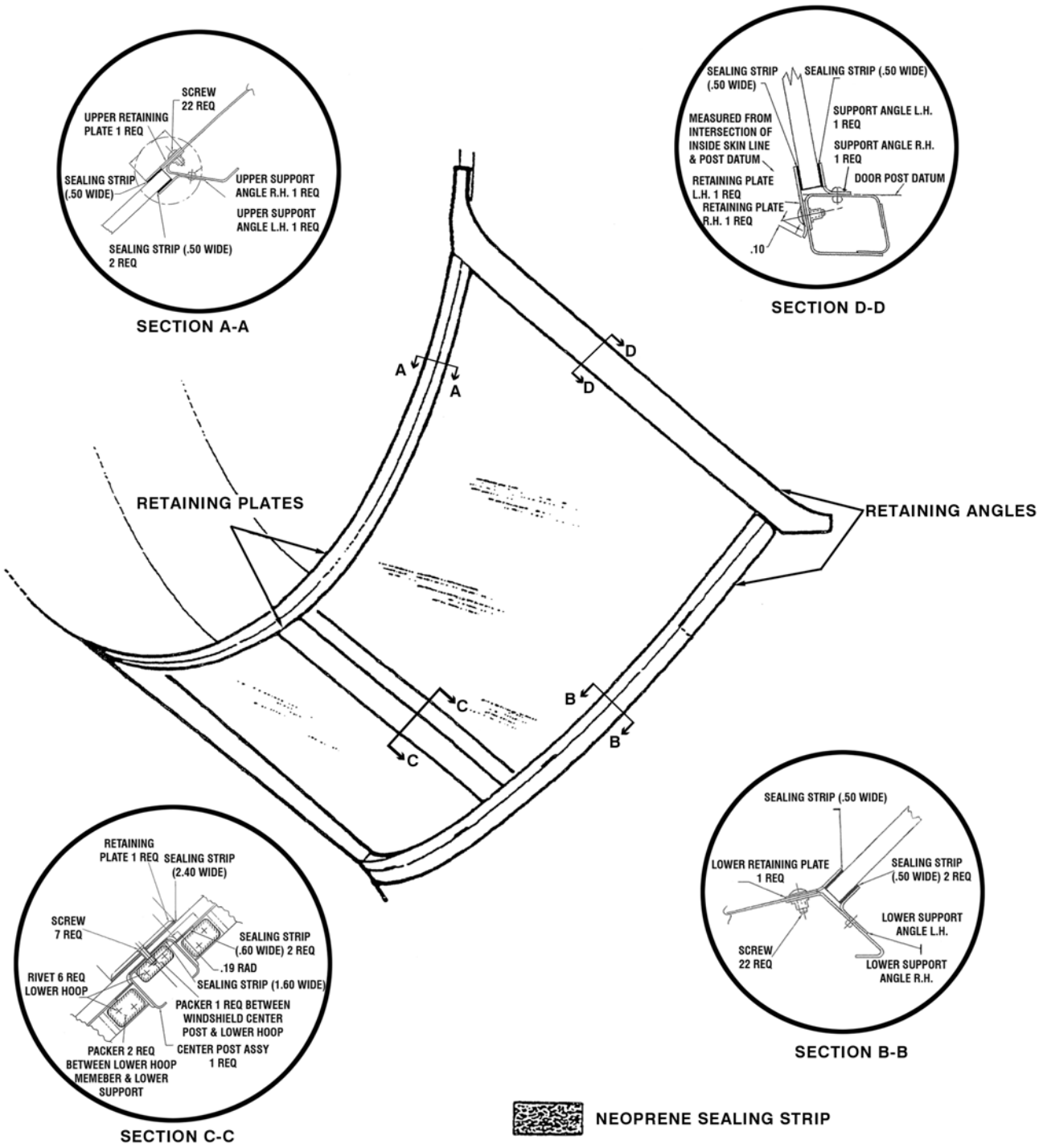
Refer to Figure 1 for details.

A. Remove Windshield

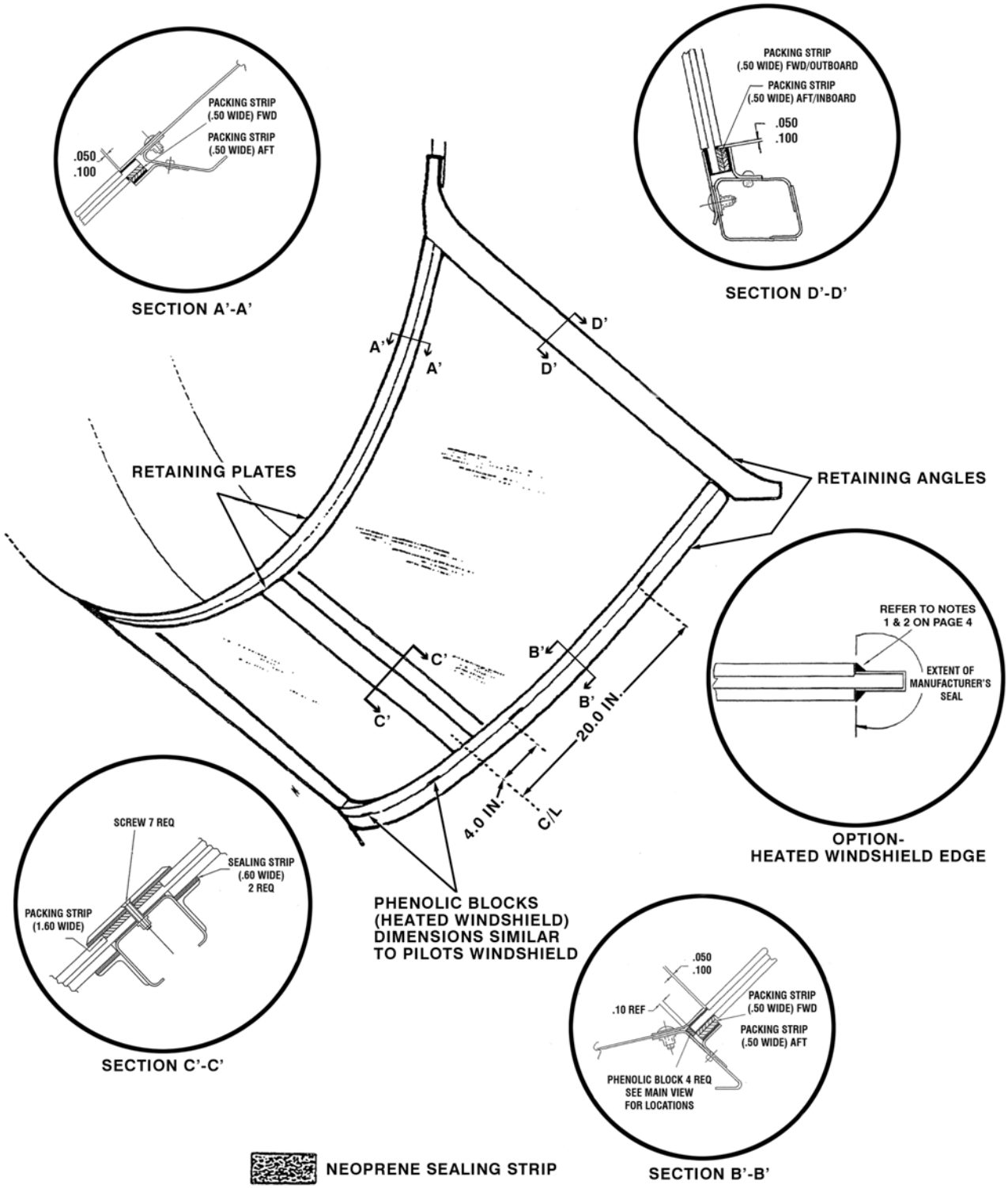
1) Disconnect windshield heat wiring at terminals (electrically heated windshield only, refer to 30-40-00, Windshield Heating System (Mod S.O.O. 6187 and Mod 6/2045) – General Data).

2) Remove screws securing retaining plates and angles.

3) Remove windshield. On electrically heated windshields only, mark position of the two phenolic blocks to facilitate reinstallation.




Windshield Installation – Base Installation (Mod 6/2032)
Figure 1




Windshield Installation – Option (S.O.O. 6187 Mod 6/2045)
Figure 2

B. Install Windshield

Note Electrically heated windshields are supplied with sealing strips and packing strips; standard windshields are supplied only with sealing strips.

1) On electrically heated windshield, check condition of sealant fillets on windshield.

Note 1. Prior to installation check that edge sealing as supplied by manufacturer is intact (refer to Figure 1 and Figure 2). If seal is broken, inspect to ensure no delamination has occurred and repair seal using PR1776M–B2 sealant.

Note 2. Prior to installation apply a fillet of PR1776M–B2 sealant to joint between glass and vinyl laminations (refer to Figure 1 and Figure 2), seal inside and outside joints around full perimeter of windshield and allow to cure for a minimum of 2 hours.

**WARNING ADEQUATE VENTILATION MUST BE PROVIDED AND THE USE OF PROTECTIVE CLOTHING STRICTLY ADHERED TO AT ALL TIMES DUE TO THE TOXIC NATURE OF THE SOLVENTS.**

2) Clean off any old sealant on windshield structure, retaining plates and retaining angles using a non-metallic scraper and naphtha.

3) On electrically heated windshields, position phenolic blocks at base of windshield structure.

4) Prepare adhesive and sealants as follows:

a) In a clean, wax free container, mix 10 parts by weight of Part A base PR1428–B2 with 100 parts by weight of Part B. The resulting sealant will have an application time of 2 hours and a full cure time of 36 hours at 75°F and 50% R.H.

b) In a clean, wax free container, mix 10 parts by weight of Part A base PR1776M–B2 with 100 parts by weight of Part B. The resulting sealant will have an application time of 2 hours and a full cure of 72 hours at 75°F and 50% R.H.

c) Thoroughly agitate the EC1357N adhesive in its own container, ensuring that all solids are dispersed by scraping the bottom of the container with a suitable tool.

5) For electrically heated windshields, bond pairs of packing strips, -55 and -57, -59 and -61, -63 and -65 together as follows using EC1357N:

a) Apply a thin uniform coat of adhesive to both bonding surfaces.

b) Allow to air dry until it is tacky, but does not transfer to the finger when touched lightly.

c) Press the parts together and roll down using a rubber or stitch roller to ensure intimate contact over the full bonding surface.

6) Apply two thin coats of release agent (parting Agent 13, Alpha/Owens Corning) to the faying surfaces of the windshield frame structure. Allow each coat to touch dry between coats and before assembly or applying sealant.

7) Apply PR1428–B2 sealant to windshield frame structure. Ensure that sufficient sealant is applied to fill gaps. Seal all gaps in excess of 0.10 inch with 2 plies of aluminum foil tape (3M 425 or Mystic 7453). Immediately install the windshield together with the sealing or packing strips.

8) Apply PR1428–B2 sealant to the retaining strips and angles and immediately install them ensuring that sufficient sealant is applied to seal all screw holes. Smooth extruded material into a fillet. Install screws.

9) Retighten screws after one hour and again after two hours. Sealant will be cured sufficiently after a further two hours to permit flight.

10) Peel or scrape off excess sealant. Do not use solvent near windshield seal.

11) On electrically heated windshields, connect electrical wiring. Function test windshield heating system (refer to 30-40-00, Windshield Heating System (Mod S.O.O. 6187 and Mod 6/2045) – General Data, Para 2.A).

12) Install windshield wiper arm and blade (if removed, refer to 30-45-12, Windshield Wiper Arm and Blade (Mod 6/2088) – Maintenance Practices, Para 2.B).

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